

SHENG WU

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EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) 09/2024 - 07/2026

M.Phil in Artificial Intelligence (GPA: 3.82/4.0)

- **Core Skills & Modules:** Convex & Non-convex Optimization I, Reinforcement Learning and Optimization, Multi-modal Artificial Intelligence.
- **Member of Deep Interdisciplinary Intelligence Lab**

University of Liverpool (UoL)/Xi'an Jiaotong-Liverpool University (XJTLU) Dual Degree 09/2020 - 06/2024

BSc in Mathematics with Finance (GPA: 63/100)

- **Core Skills & Modules:** Fin. & Act. Models in R, Linear Statistical Models (R), Numerical Methods (Python & Maple); Introduction to the Methods of Applied Mathematics (MatLab); Statistics and Probability (R); Metric Spaces And Calculus; Operational Research: Probabilistic Model; Real Analysis for Financial Mathematics, Applied Probability.
- **Director of Publicity Division, Student Union** (2020.09 - 2022.05)

PUBLICATION

MVCNet: A Vision Transformer-Based Network for Multi-Label Zero-Shot Learning in Medical Imaging 06/2024

Wu, S., Gan, H-S., Lo, Y-T., Ramlee, M. H., and Basaruddin, H., in the 4th IEEE Conference Conference on Software Engineering Artificial Intelligence (SEAI 2024)

- The paper introduces MVCNet, a novel network that leverages Vision Transformers and BioBert for multi-label zero-shot learning in medical imaging, which attempts to address the challenges of incomplete disease coverage, and optimizes the diagnostic process for radiologists.

A Multi-Modal Knowledge Network for Predicting Both Seen and Unseen Classes in Medical Imaging 01/2025

Xu, W., Gan, H-S., Wu, S., Wang, Z., Ramlee, M. H., and Hafizah, W. M., in the 28th IEEE International Conference on Computer Supported Cooperative Work in Design (CSCWD 2025)

- This paper proposes MMKNet, a multi-modal network for classifying seen and unseen medical image labels using Vision Transformers and BioMedBERT with prompt tuning. It outperforms baselines on multiple datasets, aiding rare disease diagnosis.

RESEARCH EXPERIENCE

XJTLU Chat Assistant: ChatGLM-based deployment using LangChain and proprietary knowledge base 06/2023 - 08/2023

Summer Undergraduate Research Fellow Leader

XJTLU

- Conceptualized and created a customized chatbot by injecting a local database into a large language model, covering several technical processes such as text segmentation, vectorization, identifying topK most similar vectors within the text vectors, and generating answers.
- Leveraged the WebUI for knowledgebase configuration such as dependencies installation and parameter modifications.
- Fostered flexibility by enabling command-line configuration of the knowledgebase, along which a LocalDocQA module was implemented to initiate the configuration of both LLM and Embedding models.

Natural Language Processing Research Study Group 09/2021 - 06/2023

Lab Member

XJTLU

- Advanced proficiency in a variety of neural network models such as CNN and RNN, along with a solid understanding of Transformer and BERT in the NLP domain.
- Mastered Sentence Bert, CoSENT, SimCSE for focused analysis on sentence vector embedding, validated by conducting weekly independent reviews of related research papers, thereby stimulating intellectual discussions within the team.

Patent Classification Competition, China Computer Federation Patsnap 10/2022 - 11/2022

Primary Analyst

Remote

- Championed an intensive Chinese small-sample classification project utilizing Wisdom Sprout's anonymized dataset of over 900 patents, dissecting various elements such as patent ID, description, holder, and category labels.
- Independently executed Mengzi's baseline pre-training model and collaboratively enriched the raw data via innovative techniques including translation, random deletion or swapping, and synonym replacement.
- Implemented strategic enhancements to augment model accuracy through pseudo-labels, voting fusion, and parameter tuning, which culminated in a model accuracy rate of 61.59%, propelling our team into the Top 40 nationwide.

Paper category prediction with robust fusion-enhanced graph neural network 06/2022 - 08/2022

Summer Undergraduate Research Fellow

XJTLU

- Collaborated on a project utilizing Graph Neural Networks (GNNs) to predict categories of Computer Science papers on arXiv. Addressed challenges of overfitting and out-of-distribution nodes in large-scale datasets by implementing Free Large-scale Adversarial Augmentation on Graphs (FLAG) and fusing multiple GNN backbones.
- Applied DeepWalk and Graph Attention Networks (GAT) for feature representation, followed by GraphSAGE and multi-head attention fusion for enhanced node predictions. The model achieved a top 15 ranking in a node property prediction task.

PROFESSIONAL EXPERIENCE

Sucheng-Hui Private Equity Fund Management Co., Ltd.

07/2023 - 09/2023

Intern

Suzhou, China

- Developed and implemented data processing scripts using the VNPY framework and TuShare data source, significantly enhancing efficiency by automating pre-market, intra-market, and post-market data handling tasks.
- Played a key role in collecting and analyzing market data, providing crucial insights for fund management decisions.
- Initiated comprehensive industry research to support fund manager's strategic planning.

JobReady

01/2021 - 03/2022

Senior Global Campus Partner

Shanghai, China

- Spearheaded collaborations in Jiangsu province and its surrounding areas, negotiating partnerships with 10+ educational institutions, resulting in a successful alliance between JobReady and Xi'an Jiaotong-Liverpool University in early 2022.
- Contributed to the creation and evaluation of virtual work experience initiatives, involving relevant project design and development for key clients (e.g., ZhenFund and Tongfang Securities).
- Restructured and digitized projects for high-profile clients Yixin Wealth and Adobe, some of which have been launched on their website.

RallyCry Ventures

03/2021 - 04/2021

Financial Data Analyst

Remote (Boston)

- Orchestrated targeted market research initiatives, leading to the discovery and evaluation of 10+ forward-thinking startups originating from accelerators, incubators, and universities, identifying the most potential one worthy of investment.
- Streamlined data management using OpenRefine for data cleansing along with advanced data analysis and visualization using Excel, which was then integrated into a comprehensive presentation supported by analytical models such as SWOT.

Soochow Securities

01/2021 - 02/2021

Intern, Institutional Business Department

Suzhou, China

- Assisted in planning the Private Equity Fund roadshow meetings and organizing in-depth review of documents including due diligence reports, promotion briefings, and risk disclosure statements.
- Established connections with private fund managers to comprehend their needs accurately and efficiently disseminate relevant information across regional sales departments, to implement flexible approaches to accommodate diverse requests.
- Resolved inquiries from branch customer managers related to private equity products and sales processes.